

COMPOSITE MATERIAL DATASHEETS CFRP G0904/EPO52-PW

MATERIAL IDENTIFICATION

EU 1E001 - This data is controlled under EU Dual Use Regulation 2021/821, as amended. Authorization may be required for the export of the data outside of the EU.

Short informal name	G0904/EPO52-PW
Commercial name	Reinforcement: Hexcel Hexforce G0904 D 1070 TCT Matrix: Hunstman EPOCAST52 A/B
Specifications	Reinforcement: [Airbus] AIMS08-04-001-01 (, [CASA] Z-17.238) Matrix: [Airbus] AIMS08-01-002-01 (, [CASA] Z-18.177)
Description	Carbon Fiber Reinforced material (plain weave fabric, 193 g/m ²) in epoxy matrix, for structural repairs with Wet-Lay up, Vertical Bleed Process, curing at 93°C - 120°C
Version	1.0

REFERENCES

- TE-T-NT-230041 Is. A "DERIVATION OF MATERIAL DESIGN VALUES FOR CFRP PLAIN-WEAVE FABRIC G0904/EPO52 FOR WET LAY-UP REPAIRS"
- TEA-A4-NT-210317 Is. B "A400M COWLINGS. BONDED OUT-OF-AUTOCCLAVE REPAIRS. TEST RESULTS ANALYSIS."

PHYSICAL PROPERTIES

Cured Ply Thickness (CPT)	0.20 mm
Density	1500 kg/m ³
Tg-onset (wet)	>120 °C
Max. Operational Limit (MOL)	120 °C

Coefficients of Thermal Expansion CTE (1/°C)	Range	-55...20°C	20...70°C	70...120°C
α_1		3.4E-6	3.4E-6	3.4E-6
α_2		3.4E-6	3.4E-6	3.4E-6
α_3		3.4E-5	3.4E-5	3.4E-5

ELASTIC MODULI

Property Condition \ (Units)	E_1 (MPa)	E_2 (MPa)	G_{12} (MPa)	ν_{12} -	K_B -
T ≤ 80°C	60000	60000	2900	0.04	0.85
80°C < T ≤ 120°C	57000	57000	1100	0.025	

Remarks:

- The K_B is the buckling correction factor for moduli to apply to E_1 , E_2 and G_{12} in buckling stability analyses

PLAIN STRENGTHS

Ultimate stresses of the ply (from UD-laminates testing).

Property	Mean Values (MPa)				B-basis K_B	E-KDF			B-basis Values (MPa)			
	-55C/Dry	RT/Dry	80°C/W	120°C/W		-55C/Dry	80°C/W	120°C/W	-55C/Dry	RT/Dry	80°C/W	120°C/W
F_{1t}	664	735	725	640	0.83	0.90	0.99	0.87	551	610	602	531
F_{1c}	531	541	426	262	0.83	0.98	0.79	0.48	441	449	354	217
F_{2t}	664	735	725	640	0.83	0.90	0.99	0.87	551	610	602	531
F_{2c}	531	541	426	262	0.83	0.98	0.79	0.48	441	449	354	217
F_{12}	100	84	58	43	0.83	>1	0.69	0.51	83	70	48	36

Design values for checking the laminate strength using the modified Maximum Strain Criterion.

Property	Mean Values (μ ϵ)				B-basis K_B	E-KDF			B-basis Values (μ ϵ)			
	-55C/Dry	RT/Dry	80°C/W	120°C/W		-55C/Dry	80°C/W	120°C/W	-55C/Dry	RT/Dry	80°C/W	120°C/W
ϵ_{tall}	11000	12000	12000	10000	0.89	0.91	1.0	0.83	9800	10000	10000	9000
ϵ_{call}	9000	9000	7100	4300	0.83	1.0	0.78	0.47	7500	7500 5800 (*)	5900 4600 (*)	3600

Marked with (*) are cut-off values for the repair laminate in sandwich faces

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INTERLAMINAR PLY STRENGTHS

Property	Mean Values				B-basis K_B	E-KDF			B-basis Values			
	-55C/Dry	RT/Dry	80°C/W	120°C/W		-55C/Dry	80°C/W	120°C/W	-55C/Dry	RT/Dry	80°C/W	120°C/W
F_{13} (MPa)	51	47	34	23	0.83	>1	0.72	0.49	42	39	28	19
F_{3t} (MPa)	-	-	-	-	-	-	-	-	-	-	-	-
G_{IC} (N/mm)	-	-	-	-	-	-	-	-	-	-	-	-
G_{IIC} (N/mm)	-	-	-	-	-	-	-	-	-	-	-	-

OPEN HOLE AND BOLTED JOINTS

Reference material strengths in open-hole, filled-hole and bearing for the quasi-isotropic laminate.

Property	Mean Values (MPa)				B-basis K_B	E-KDF			B-basis Values (MPa)			
	-55C/Dry	RT/Dry	80°C/W	120°C/W		-55C/Dry	80°C/W	120°C/W	-55C/Dry	RT/Dry	80°C/W	120°C/W
OHT_{QI}	292	316	317	305	0.89	0.92	1.00	0.97	260	281	282	271
OHC_{QI}	329	267	204	153	0.83	>1	0.76	0.57	273	222	169	127
FHT_{QI}	293	324	-	-	0.89	0.90	-	-	261	288	-	-
FHC_{QI}	-	399	323	256	0.83	-	0.81	0.64	-	331	268	212
FBR_{QI}	781	939	762	623	0.83	0.83	0.81	0.66	648	779	632	517
F_{PO} (*)	292	316	317	305	0.89	0.92	1.00	0.97	260	281	282	271

(*) Pull-out strength under tension load:

Allow. stress: $\tau_{PO} = EKDF \cdot K_B \cdot F_{PO} \cdot \Psi(t/d)$... with t thickness, d shank diameter, $\Psi = 1 + 0.6 \cdot (t/d - 1)$ if $t/d < 1$, otherwise $\Psi = 1$

Allow. force: $T_{PO} = \tau_{PO} \cdot \pi \cdot D_h \cdot t$... with t , again, the laminate thickness, and D_h the diameter of the bolt head

For bolted joints analysis at 120°C/W, under in-plane bypass/bearing loads, Airbus DS software *URCO* shall be used, selecting material "Group IV Fabric", and applying the following KDF to *URCO* allowables:

Property	Group IV - URCO			G0904/EPO52 at 120°C/W	Correction to URCO, "Fabric Group IV"	
	E_x (MPa)	ε_{bp} (με)	σ (MPa)	σ (MPa)	Allowable	$KDF@120^\circ C$
FHT	35082	5081	178	271 (**)	A, B	1.0
FHC		-5081	178	212	D	1.0
FBR	-	-	660	517	C	0.78

(**) FHT for G0924/EPO52 material assumed equal to OHT

REPAIR MATERIAL AS ADHESIVE

Design data to use with Airbus DS software *ARES* (see further details in TEA-A4-NT-210317 Is. B).

Property	cpt	G	τ_{lin}	τ_u	γ_u	E	F_{tu}
Cond. \ Units	(mm)	(MPa)	(MPa)	(MPa)	-	(MPa)	(με)
RT/Dry/AR	0.2	1000	20	40	0.1	2700	29
80°C/Wet		600	14	20	0.4	1600	24
120°C/Wet		100	5	10	0.8	270	17

COMPOSITE DAMAGE TOLERANCE

Reserved for future versions.